

16 The activity of an enzyme can be affected by a competitive inhibitor.

Which row is correct for the effect of a competitive inhibitor on the value of K_m and the reason for this?

	effect of a competitive inhibitor on the value of K_m	reason
A	increased	Few substrate molecules will bind to the active site when the substrate concentration is low because the active site is blocked by the inhibitor.
B	increased	V_{max} will increase because the substrate will only bind to the active site when the concentration of the substrate is high.
C	no effect	At high substrate concentrations, substrate molecules are still able to bind to the enzymes in the presence of a competitive inhibitor.
D	no effect	The inhibitor molecule does not bind to the active site; it binds to a site on another part of the enzyme.